EXAMPLE 9.1

The radius of the 5th orbit of hydrogen atom is 13.25 Å. Calculate the de broglie wavelength of the electron orbitting in the 5th orbit.

Solution:

$2\pi r = n\lambda$

$2 \times 3.14 \times 13.25\text{Å} = 5 \times \lambda$

$\therefore \lambda = 16.64\text{Å}$

EXAMPLE 9.2

Find the (i) angular momentum (ii) velocity of the electron revolving in the 5th orbit of hydrogen atom.

($h = 6.6 \times 10^{-34}$ Js, $m = 9.1 \times 10^{-31}$ kg)

Solution

(i) Angular momentum is given by

$$l = n\hbar = \frac{nh}{2\pi}$$

$$= \frac{5 \times 6.6 \times 10^{-34}}{2 \times 3.14} = 5.25 \times 10^{-34}\ \text{kgm}^2\text{s}^{-1}$$

(ii) Velocity is given by

$$\text{Velocity } \upsilon = \frac{l}{mr}$$

$$= \frac{(5.25 \times 10^{-34}\ \cancel{\text{kgm}^2}\text{s}^{-1})}{(9.1 \times 10^{-31}\ \cancel{\text{kg}})(13.25 \times 10^{-10}\ \cancel{\text{m}})}$$

$$\upsilon = 4.4 \times 10^5\ \text{ms}^{-1}$$

Radius of the orbit of the electron and velocity of the electron

Consider an atom which contains the nucleus at rest and an electron revolving around the nucleus in a circular orbit of radius r_n as shown in Figure 9.17. Nucleus is made up of protons and neutrons. Since proton is positively charged and neutron is electrically neutral, the charge of a nucleus is entirely due to the charge of protons.

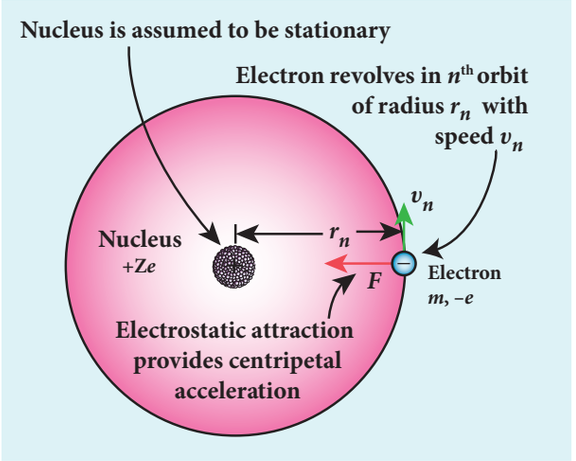


Figure 9.17 Electron revolving around the nucleus

Let Z be the atomic number of the atom, then $+Ze$ is the charge of the nucleus. Let $-e$ be the charge of the electron. From Coulomb's law, the force of attraction between the nucleus and the electron is

$$\vec{F}_{coloumb} = \frac{1}{4\pi\varepsilon_0}\frac{(+Ze)(-e)}{r_n^2}\hat{\text{r}}$$

$$= -\frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r_n^2}\hat{\text{r}}$$

This force provides necessary centripetal force

$$\vec{F}_{centripetal} = \frac{m\upsilon_n^2}{r_n}\hat{\text{r}}$$

where m be the mass of the electron that moves with a velocity v_n in a circular orbit. Therefore,

$$\left|\vec{F}_{coloumb}\right| = \left|\vec{F}_{centripetal}\right|$$

$$\frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r_n^2} = \frac{m\upsilon_n^2}{r_n}$$

Multiplied and divided by 'm'

$$r_n = \frac{4\pi\varepsilon_0(m\upsilon_n r_n)^2}{Zme^2} \qquad (9.15)$$

From Bohr's assumption, the angular momentum quantization condition, $m\upsilon_n r_n = l_n = n\hbar$,

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